KERALA DIPLOMA CURRICULUM REVISION 2026

Course 3072 | 3 Credits: 3 | Program Core | Source-grounded

Mechanical Operations

 To impart thorough knowledge in mechanical operations used in chemical plants.

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTR syllabus PDF for Course 3072 is the authoritative source for course information, objectives, course outcomes, CO–PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 3072.pdf from the uploaded official SITTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Chemical Engineering
Course code	3072
Course title	Mechanical Operations
Semester	3 Credits: 3
Course category	Program Core
Periods per week	3 (L: 2 T: 1 P: 0) Periods per semester: 45
Periods per semester	45
Credits	3
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

16 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD

How to use this lesson

First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- To impart thorough knowledge in mechanical operations used in chemical plants.
- To impart operational skills for chemical plant operations.

Course outcomes

CO	Official outcome	Study level
CO1	Summarise size reduction and size separation. 9 Understanding	See official syllabus
CO2	Interpret solid - liquid separation equipments. 14 Applying Summarise the industrial operations agitation	See official syllabus
CO3	14 Understanding and mixing. Summarise the handling of solids in chemical	See official syllabus
CO4	8 Understanding process industries.	See official syllabus

CO–PO mapping evidence

Row	Extracted official mapping
CO1	CO1 2
CO2	CO2 3
CO3	CO3 3
CO4	CO4 2

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	Summarise size reduction and size separation.	5	As specified
Module 2	Interpret solid - liquid separation equipments.	4	As specified
Module 3	Summarise the industrial operations agitation and mixing.	4	As specified
Module 4	Summarise the handling of solids in chemical process industries.	3	As specified

MODULE 1

Summarise size reduction and size separation.

This module is presented from the official syllabus scope for Mechanical Operations. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Summarise size reduction and size separation.
- Official duration: As specified
- Official topic entries captured: 5
- The full source excerpt is preserved below for coverage checking.

▼ Comprehend the properties of particles. 2 Understanding.

Comprehend the properties of particles. 2 Understanding. is an official topic in Module 1 of Mechanical Operations. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Comprehend the properties of particles. 2 Understanding. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, comprehend the properties of particles. 2 understanding. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Comprehend the properties of particles. 2 Understanding. means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ Comprehend the properties of particles. 2 Understanding. □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

▼ Summarise size reduction. 1 Understanding.

Summarise size reduction. 1 Understanding. is an official topic in Module 1 of Mechanical Operations. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Summarise size reduction. 1 Understanding. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, summarise size reduction. 1 understanding. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Summarise size reduction. 1 Understanding. means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ Summarise size reduction. 1 Understanding. □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

▼ Illustrate size reduction equipment. 3 Understanding.

Illustrate size reduction equipment. 3 Understanding. is an official topic in Module 1 of Mechanical Operations. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Illustrate size reduction equipment. 3 Understanding. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, illustrate size reduction equipment. 3 understanding. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Illustrate size reduction equipment. 3 Understanding. means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-[illegible] Illustrate size reduction equipment. 3 Understanding. [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

▾ Comprehend the theories behind crushing. 1 Understanding.

Comprehend the theories behind crushing. 1 Understanding. is an official topic in Module 1 of Mechanical Operations. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Comprehend the theories behind crushing. 1 Understanding. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, comprehend the theories behind crushing. 1 understanding. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare

Aspect	What to prepare
Definition/identity	What Comprehend the theories behind crushing. 1 Understanding. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ Comprehend the theories behind crushing. 1 Understanding. □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

▼ Summarise the methods of size separation. 2 Understanding. Define: Particle shape, particle size, specific surface area of the mixture, average particle size, and shape factor. Size reduction: List the nature of materials to be crushed - List the classification of crushing and grinding equipment. Differentiate closed circuit grinding and open circuit grinding. Illustrate the working of: Blake Jaw crusher - Gyratory crusher -Single Roll crusher and Smooth roll crusher - ball mill and tube mill Explain the theory behind the action of crushing rolls - Define the angle of Nip of crushing rolls. Laws of crushing- State Rittinger's law, Kick's law and Bond's law. Size separation- Compare the Tyler and U.S standard screens. List the types of screening equipment. Illustrate the working of grizzlies, trommel, vibrating screens and magnetic separator. Differentiate shaking and vibrating screen - Ideal and actual screen - Effectiveness of screen

Summarise the methods of size separation. 2 Understanding. Define: Particle shape, particle size, specific surface area of the mixture, average particle size, and shape factor. Size reduction: List the nature of materials to be crushed - List the classification of crushing and grinding equipment. Differentiate closed circuit grinding and open circuit grinding. Illustrate the working of: Blake Jaw crusher - Gyratory crusher -Single Roll crusher and Smooth roll crusher - ball mill and tube mill Explain the theory behind the action of crushing rolls - Define the angle of Nip of crushing rolls. Laws of crushing- State Rittinger's law, Kick's law and Bond's law. Size separation- Compare the Tyler and U.S standard screens. List the types of screening equipment. Illustrate the working of grizzlies, trommel, vibrating screens and magnetic separator. Differentiate shaking and vibrating screen - Ideal and actual screen - Effectiveness of screen is an official topic in Module 1 of Mechanical Operations. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Summarise the methods of size separation. 2 Understanding. Define: Particle shape, particle size, specific surface area of the mixture, average particle size, and shape factor. Size reduction: List the nature of materials to be crushed - List the classification of crushing and grinding equipment. Differentiate closed circuit grinding and open circuit grinding. Illustrate the working of: Blake Jaw crusher - Gyratory crusher - Single Roll crusher and Smooth roll crusher - ball mill and tube mill Explain the theory behind the action of crushing rolls - Define the angle of Nip of crushing rolls. Laws of crushing- State Rittinger’s law, Kick’s law and Bond’s law. Size separation- Compare the Tyler and U.S standard screens. List the types of screening equipment. Illustrate the working of grizzlies, trommel, vibrating screens and magnetic separator. Differentiate shaking and vibrating screen - Ideal and actual screen - Effectiveness of screen using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, summarise the methods of size separation. 2 understanding. define: particle shape, particle size, specific surface area of the mixture, average particle size, and shape factor. size reduction: list the nature of materials to be crushed - list the classification of crushing and grinding equipment. differentiate closed circuit grinding and open circuit grinding. illustrate the working of: blake jaw crusher - gyratory crusher -single roll crusher and smooth roll crusher - ball mill and tube mill explain the theory behind the action of crushing rolls - define the angle of nip of crushing rolls. laws of crushing- state rittinger’s law, kick’s law and bond’s law. size separation- compare the tyler and u.s standard screens. list the types of screening equipment. illustrate the working of grizzlies, trommel, vibrating screens and magnetic separator. differentiate shaking and vibrating screen - ideal and actual screen - effectiveness of screen should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Summarise the methods of size separation. 2 Understanding. Define: Particle shape, particle size, specific surface area of the mixture, average particle size, and shape factor. Size reduction: List the nature of materials to be crushed - List the classification of crushing and grinding equipment. Differentiate closed circuit grinding and open circuit grinding. Illustrate the working of: Blake Jaw crusher - Gyratory crusher - Single Roll crusher and Smooth roll crusher - ball mill and tube mill Explain the theory behind the action of crushing rolls - Define the angle of Nip of crushing rolls. Laws of crushing- State Rittinger’s law, Kick’s law and Bond’s law. Size separation- Compare the Tyler and U.S standard screens. List the types of screening equipment. Illustrate the working of grizzlies, trommel, vibrating screens and magnetic separator. Differentiate shaking and vibrating screen - Ideal and actual screen - Effectiveness of screen means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ Summarise the methods of size separation. 2 Understanding. Define: Particle shape, particle size, specific surface area of the mixture, average particle size, and shape factor. Size reduction: List the nature of materials to be crushed - List the classification of crushing and grinding equipment. Differentiate closed circuit grinding and open circuit grinding. Illustrate the working of: Blake Jaw crusher - Gyratory crusher -Single Roll crusher and Smooth roll crusher - ball mill and tube mill Explain the theory behind the action of crushing rolls - Define the angle of Nip of crushing rolls. Laws of crushing- State Rittinger’s law, Kick’s law and Bond’s law. Size separation- Compare the Tyler and U.S standard screens. List the types of screening equipment. Illustrate the working of grizzlies, trommel, vibrating screens and magnetic separator. Differentiate shaking and vibrating screen - Ideal and actual screen - Effectiveness of screen □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

▶ **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

Interpret solid - liquid separation equipments.

This module is presented from the official syllabus scope for Mechanical Operations. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Interpret solid - liquid separation equipments.
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

▼ Comprehend the principles of settling.

Comprehend the principles of settling. is an official topic in Module 2 of Mechanical Operations. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Comprehend the principles of settling. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, comprehend the principles of settling. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Comprehend the principles of settling. means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ Comprehend the principles of settling. □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

▼ Interpret sedimentation processes.

Interpret sedimentation processes. is an official topic in Module 2 of Mechanical Operations. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Interpret sedimentation processes. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, interpret sedimentation processes. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Interpret sedimentation processes. means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ Interpret sedimentation processes. □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

▼ Interpret industrial filtration operations.

Interpret industrial filtration operations. is an official topic in Module 2 of Mechanical Operations. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Interpret industrial filtration operations. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, interpret industrial filtration operations. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Interpret industrial filtration operations. means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ Interpret industrial filtration operations. □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

▼ Comprehend centrifuging operation. 2 Understanding. Settling: Types of settling - Stokes law – Define terminal settling velocity. Sedimentation - Batch and continuous sedimentation - Kynch theory – Estimation of thickener area - Illustrate continuous thickener -Illustrate Double cone classifier - Illustrate Tabling, Jigging and Froth floatation. Filtration –Classify filtration equipment based on external force, operation, and nature of filtration - Illustrate the working of sand filter, washing and non-washing plate and frame filter press, leaf filter and Rotary Drum Filter - List the factors affecting rate of filtration - Role of filter aids in filtration - List the commonly used filter aids - Differentiate compressible and non-compressible filter cakes - Differentiate constant pressure filtration and constant volume filtration. Centrifugation: List the classification of centrifuges based on operation and the drive position. Illustrate the working top driven centrifuge, bottom driven centrifuge, disc centrifuge and super centrifuge.

Comprehend centrifuging operation. 2 Understanding. Settling: Types of settling - Stokes law – Define terminal settling velocity. Sedimentation - Batch and continuous sedimentation - Kynch theory – Estimation of thickener area - Illustrate continuous thickener -Illustrate Double cone classifier - Illustrate Tabling, Jigging and Froth floatation. Filtration –Classify filtration equipment based on external force, operation, and nature of filtration - Illustrate the working of sand filter, washing and non-washing plate and frame filter press, leaf filter and Rotary Drum Filter - List the factors affecting rate of filtration - Role of filter aids in filtration - List the commonly used filter aids - Differentiate compressible and non-compressible filter cakes - Differentiate constant pressure filtration and constant volume filtration. Centrifugation: List the classification of centrifuges based on operation and the drive position. Illustrate the working top driven centrifuge, bottom driven centrifuge, disc centrifuge and super centrifuge. is an official topic in Module 2 of Mechanical Operations. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Comprehend centrifuging operation. 2 Understanding. Settling: Types of settling - Stokes law – Define terminal settling velocity. Sedimentation - Batch and continuous sedimentation - Kynch theory – Estimation of thickener area - Illustrate continuous thickener -Illustrate Double cone classifier - Illustrate Tabling, Jigging and Froth floatation. Filtration –Classify filtration equipment based on external force, operation, and nature of filtration - Illustrate the working of sand filter, washing and non-washing plate and frame filter press, leaf filter and Rotary Drum Filter - List the factors affecting rate of filtration - Role of filter aids in filtration - List the commonly used filter aids - Differentiate compressible and non-compressible filter cakes - Differentiate constant pressure filtration and constant volume filtration. Centrifugation: List the classification of centrifuges based on operation and the drive position. Illustrate the working top driven centrifuge, bottom driven centrifuge, disc centrifuge and super centrifuge. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, comprehend centrifuging operation. 2 understanding. settling: types of settling - stokes law – define terminal settling velocity. sedimentation - batch and continuous sedimentation - kynch theory – estimation of thickener area - illustrate continuous thickener -illustrate double cone classifier - illustrate tabling, jigging and froth floatation. filtration –classify filtration equipment based on external force, operation, and nature of filtration - illustrate the working of sand filter, washing and non-washing plate and frame filter press, leaf filter and rotary drum filter - list the factors affecting rate of filtration - role of filter aids in filtration - list the commonly used filter aids - differentiate compressible and non-compressible filter cakes - differentiate constant pressure filtration and constant volume filtration. centrifugation: list the classification of centrifuges based on operation and the drive position. illustrate the working top driven centrifuge, bottom driven centrifuge, disc centrifuge and super centrifuge. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Comprehend centrifuging operation. 2 Understanding. Settling: Types of settling - Stokes law – Define terminal settling velocity. Sedimentation - Batch and continuous sedimentation - Kynch theory – Estimation of thickener area - Illustrate continuous thickener -Illustrate Double cone classifier - Illustrate Tabling, Jigging and Froth floatation. Filtration –Classify filtration equipment based on external force, operation, and nature of filtration - Illustrate the working of sand filter, washing and non-washing plate and frame filter press, leaf filter and Rotary Drum Filter - List the factors affecting rate of filtration - Role of filter aids in filtration - List the commonly used filter aids - Differentiate compressible and non-compressible filter cakes - Differentiate constant pressure filtration and constant volume filtration. Centrifugation: List the classification of centrifuges based on operation and the drive position. Illustrate the working top driven centrifuge, bottom driven centrifuge, disc centrifuge and super centrifuge. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ Comprehend centrifuging operation. 2 Understanding. Settling: Types of settling - Stokes law – Define terminal settling velocity. Sedimentation - Batch and continuous sedimentation - Kynch theory – Estimation of thickener area - Illustrate continuous thickener -Illustrate Double cone classifier - Illustrate Tabling, Jigging and Froth floatation. Filtration –Classify filtration equipment based on external force, operation, and nature of filtration - Illustrate the working of sand filter, washing and non-washing plate and frame filter press, leaf filter and Rotary Drum Filter - List the factors affecting rate of filtration - Role of filter aids in filtration - List the commonly used filter aids - Differentiate compressible and non-compressible filter cakes - Differentiate constant pressure filtration and constant volume filtration. Centrifugation: List the classification of centrifuges based on operation and the drive position. Illustrate the working top driven centrifuge, bottom driven centrifuge, disc centrifuge and super centrifuge. □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

▶ **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

Summarise the industrial operations agitation and mixing.

This module is presented from the official syllabus scope for Mechanical Operations. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Summarise the industrial operations agitation and mixing.
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

▾ Outline agitation operation.

Outline agitation operation. is an official topic in Module 3 of Mechanical Operations. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Outline agitation operation. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, outline agitation operation. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Outline agitation operation. means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-[illegible] Outline agitation operation. [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

▼ Summarise agitation equipment. 4 Understanding Identify the dimensionless numbers related to

Summarise agitation equipment. 4 Understanding Identify the dimensionless numbers related to is an official topic in Module 3 of Mechanical Operations. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Summarise agitation equipment. 4 Understanding Identify the dimensionless numbers related to using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, summarise agitation equipment. 4 understanding identify the dimensionless numbers related to should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare

Aspect	What to prepare
Definition/identity	What Summarise agitation equipment. 4 Understanding Identify the dimensionless numbers related to means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ Summarise agitation equipment. 4 Understanding Identify the dimensionless numbers related to □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□.

▼ 4 Understanding agitation operation.

4 Understanding agitation operation. is an official topic in Module 3 of Mechanical Operations. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding agitation operation. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding agitation operation. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding agitation operation. means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ 4 Understanding agitation operation. □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□□, steps, application □□□□□□ □□□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

▾ Summarise the working of mixing equipments. 4 Understanding Agitation: Define Agitation - List the purpose of agitation - Flow pattern in agitated vessels. Agitation equipment: Illustrate agitation equipment - Types of Impellors, Sketch of propellers paddles and turbines – Identify the function of draft tube and Baffle. State: Power number, Froude Number and Reynold’s no. Mixing: Illustrate dry mixer, Banbury mixer, ‘V’ type mixer and kneading machine.

Summarise the working of mixing equipments. 4 Understanding Agitation: Define Agitation - List the purpose of agitation - Flow pattern in agitated vessels. Agitation equipment: Illustrate agitation equipment - Types of Impellors, Sketch of propellers paddles and turbines – Identify the function of draft tube and Baffle. State: Power number, Froude Number and Reynold’s no. Mixing: Illustrate dry mixer, Banbury mixer, ‘V’ type mixer and kneading machine. is an official topic in Module 3 of Mechanical Operations. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Summarise the working of mixing equipments. 4 Understanding Agitation: Define Agitation - List the purpose of agitation - Flow pattern in agitated vessels. Agitation equipment: Illustrate agitation equipment - Types of Impellors, Sketch of propellers paddles and turbines – Identify the function of draft tube and Baffle. State: Power number, Froude Number and Reynold’s no. Mixing: Illustrate dry mixer, Banbury mixer, ‘V’ type mixer and kneading machine. using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, summarise the working of mixing equipments. 4 understanding agitation: define agitation - list the purpose of agitation - flow pattern in agitated vessels. agitation equipment: illustrate agitation equipment - types of impellors, sketch of propellers paddles and turbines – identify the function of draft tube and baffle. state: power number, froude number and reynold’s no. mixing: illustrate dry mixer, banbury mixer, ‘v’ type mixer and kneading machine. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Summarise the working of mixing equipments. 4 Understanding Agitation: Define Agitation - List the purpose of agitation - Flow pattern in agitated vessels. Agitation equipment: Illustrate agitation equipment - Types of Impellors, Sketch of propellers paddles and turbines – Identify the function of draft tube and Baffle. State: Power number, Froude Number and Reynold’s no. Mixing: Illustrate dry mixer, Banbury mixer, ‘V’ type mixer and kneading machine. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ Summarise the working of mixing equipments. 4 Understanding Agitation: Define Agitation - List the purpose of agitation - Flow pattern in agitated vessels. Agitation equipment: Illustrate agitation equipment - Types of Impellors, Sketch of propellers paddles and turbines – Identify the function of draft tube and Baffle. State: Power number, Froude Number and Reynold’s no. Mixing: Illustrate dry mixer, Banbury mixer, ‘V’ type mixer and kneading machine. □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

▶ **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

Summarise the handling of solids in chemical process industries.

This module is presented from the official syllabus scope for Mechanical Operations. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Summarise the handling of solids in chemical process industries.
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

▾ Summarise the storage of solids in industries.

Summarise the storage of solids in industries. is an official topic in Module 4 of Mechanical Operations. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Summarise the storage of solids in industries. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, summarise the storage of solids in industries. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Summarise the storage of solids in industries. means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ Summarise the storage of solids in industries. □□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

▾ Illustrate mechanical conveying systems.

Illustrate mechanical conveying systems. is an official topic in Module 4 of Mechanical Operations. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Illustrate mechanical conveying systems. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, illustrate mechanical conveying systems. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Illustrate mechanical conveying systems. means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-[illegible] Illustrate mechanical conveying systems. [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

▼ Comprehend pneumatic conveying systems. 3 Understanding Storage of solids- Hoppers, bins, silos - Angle of repose - Open storage of solids. Classification of conveyors – Illustrate Belt conveyor, Apron conveyor, Screw conveyor, Bucket elevator. Illustrate Pneumatic conveying system - Illustrate the construction and working of cyclone separator, bag filter and ESP. Text / Reference T/R Book Title/Author T1 Introduction to Chemical Engineering; Walter.L.Badger & Julius.T.Banchero Mechanical operations; Anup Kumar Swain, hemalathapatra and Gopendra R1 Kishore Roy R2 Unit Operations of Chemical Engineering; P.Chathopadhyay R3 Unit Operations; Warren.L.McCabe&Julian.C.Smith, R4 Perry's Chemical Engineers' Handbook; Robert H Perry, Don W. Green Online Resources Sl.No Website Link 1 https://www.nptel.ac.in/courses/103/107/103107123/

Comprehend pneumatic conveying systems. 3 Understanding Storage of solids- Hoppers, bins, silos - Angle of repose - Open storage of solids. Classification of conveyors – Illustrate Belt conveyor, Apron conveyor, Screw conveyor, Bucket elevator. Illustrate Pneumatic conveying system - Illustrate the construction and working of cyclone separator, bag filter and ESP. Text / Reference T/R Book Title/Author T1 Introduction to Chemical Engineering; Walter.L.Badger & Julius.T.Banchero Mechanical operations; Anup Kumar Swain, hemalathapatra and Gopendra R1 Kishore Roy R2 Unit Operations of Chemical Engineering; P.Chathopadhyay R3 Unit Operations; Warren.L.McCabe&Julian.C.Smith, R4 Perry's Chemical Engineers' Handbook; Robert H Perry, Don W. Green Online Resources Sl.No Website Link 1 https://www.nptel.ac.in/courses/103/107/103107123/ is an official topic in Module 4 of Mechanical Operations. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Comprehend pneumatic conveying systems. 3 Understanding Storage of solids- Hoppers, bins, silos - Angle of repose - Open storage of solids. Classification of conveyors – Illustrate Belt conveyor, Apron conveyor, Screw conveyor, Bucket elevator. Illustrate Pneumatic conveying system - Illustrate the construction and working of cyclone separator, bag filter and ESP. Text / Reference T/R Book Title/Author T1 Introduction to Chemical Engineering; Walter.L.Badger & Julius.T.Banchero Mechanical operations; Anup Kumar Swain, hemalathapatra and Gopendra R1 Kishore Roy R2 Unit Operations of Chemical Engineering; P.Chathopadhyay R3 Unit Operations; Warren.L.McCabe&Julian.C.Smith, R4 Perry's Chemical Engineers' Handbook; Robert H Perry, Don W. Green Online Resources Sl.No Website Link 1 https://www.nptel.ac.in/courses/103/107/103107123/ using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, comprehend pneumatic conveying systems. 3 understanding storage of solids- hoppers, bins, silos - angle of repose - open storage of solids. classification of conveyors – illustrate belt conveyor, apron conveyor, screw conveyor, bucket elevator. illustrate pneumatic conveying system - illustrate the construction and working of cyclone separator, bag filter and esp. text / reference t/r book title/author t1 introduction to chemical engineering; walter.l.badger & julius.t.banchero mechanical operations; anup kumar swain, hemalathapatra and gopendra r1 kishore roy r2 unit operations of chemical engineering; p.chathopadhyay r3 unit operations; warren.l.mccabe&julian.c.smith, r4 perry's chemical engineers' handbook; robert h perry, don w. green online resources sl.no website link 1 https://www.nptel.ac.in/courses/103/107/103107123/ should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Comprehend pneumatic conveying systems. 3 Understanding Storage of solids- Hoppers, bins, silos - Angle of repose - Open storage of solids. Classification of conveyors – Illustrate Belt conveyor, Apron conveyor, Screw conveyor, Bucket elevator. Illustrate Pneumatic conveying system - Illustrate the construction and working of cyclone separator, bag filter and ESP. Text / Reference T/R Book Title/Author T1 Introduction to Chemical Engineering; Walter.L.Badger & Julius.T.Banchero Mechanical operations; Anup Kumar Swain, hemalathapatra and Gopendra R1 Kishore Roy R2 Unit Operations of Chemical Engineering; P.Chathopadhyay R3 Unit Operations; Warren.L.McCabe&Julian.C.Smith, R4 Perry's Chemical Engineers' Handbook; Robert H Perry, Don W. Green Online Resources Sl.No Website Link 1 https://www.nptel.ac.in/courses/103/107/103107123/ means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ Comprehend pneumatic conveying systems. 3 Understanding Storage of solids- Hoppers, bins, silos - Angle of repose - Open storage of solids. Classification of conveyors – Illustrate Belt conveyor, Apron conveyor, Screw conveyor, Bucket elevator. Illustrate Pneumatic conveying system - Illustrate the construction and working of cyclone separator, bag filter and ESP. Text / Reference T/R Book Title/Author T1 Introduction to Chemical Engineering; Walter.L.Badger & Julius.T.Banchero Mechanical operations; Anup Kumar Swain, hemalathapatra and Gopendra R1 Kishore Roy R2 Unit Operations of Chemical Engineering; P.Chathopadhyay R3 Unit Operations; Warren.L.McCabe&Julian.C.Smith, R4 Perry's Chemical Engineers' Handbook; Robert H Perry, Don W. Green Online Resources Sl.No Website Link 1 https://www.nptel.ac.in/courses/103/107/103107123/ □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

▶ **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

```
Definition → Principle → Components/steps → Application → Conclusion
```

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

```
Aim → Apparatus → Procedure → Observation → Result → Safety
```

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION

Diagram and source-transcript library

Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

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The official assessment structure is reproduced from the supplied PDF.
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Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

▶ **Define or explain Comprehend the properties of particles. 2 Understanding.. (3 marks)**

▶ **Define or explain Summarise size reduction. 1 Understanding.. (3 marks)**

▶ **Define or explain Illustrate size reduction equipment. 3 Understanding.. (3 marks)**

▶ **Define or explain Comprehend the theories behind crushing. 1 Understanding.. (3 marks)**

Module 2

▶ **Define or explain Comprehend the principles of settling.. (3 marks)**

▶ **Define or explain Interpret sedimentation processes.. (3 marks)**

▶ **Define or explain Interpret industrial filtration operations.. (3 marks)**

▶ **Define or explain Comprehend centrifuging operation. 2 Understanding. Settling: Types of settling - Stokes law – Define terminal settling velocity. Sedimentation - Batch and continuous sedimentation - Kynch theory – Estimation of thickener area - Illustrate continuous**

thickener -Illustrate Double cone classifier - Illustrate Tabling, Jigging and Froth floatation. Filtration –Classify filtration equipment based on external force, operation, and nature of filtration - Illustrate the working of sand filter, washing and non-washing plate and frame filter press, leaf filter and Rotary Drum Filter - List the factors affecting rate of filtration - Role of filter aids in filtration - List the commonly used filter aids - Differentiate compressible and non-compressible filter cakes - Differentiate constant pressure filtration and constant volume filtration. Centrifugation: List the classification of centrifuges based on operation and the drive position. Illustrate the working top driven centrifuge, bottom driven centrifuge, disc centrifuge and super centrifuge.. (3 marks)

Module 3

▶ **Define or explain Outline agitation operation.. (3 marks)**

▶ **Define or explain Summarise agitation equipment. 4 Understanding Identify the dimensionless numbers related to. (3 marks)**

▶ **Define or explain 4 Understanding agitation operation.. (3 marks)**

▶ **Define or explain Summarise the working of mixing equipments. 4 Understanding Agitation: Define Agitation - List the purpose of agitation - Flow pattern in agitated vessels. Agitation equipment: Illustrate agitation equipment - Types of Impellors, Sketch of propellers paddles and turbines – Identify the function of draft tube and Baffle. State: Power number, Froude Number and Reynold’s no. Mixing: Illustrate dry mixer, Banbury mixer, ‘V’ type mixer and kneading machine.. (3 marks)**

Module 4

▶ **Define or explain Summarise the storage of solids in industries.. (3 marks)**

▶ **Define or explain Illustrate mechanical conveying systems.. (3 marks)**

▶ **Define or explain Comprehend pneumatic conveying systems. 3 Understanding Storage of solids- Hoppers, bins, silos - Angle of repose - Open storage of solids. Classification of conveyors – Illustrate Belt conveyor, Apron conveyor, Screw conveyor, Bucket elevator. Illustrate Pneumatic conveying system - Illustrate the construction and working of cyclone separator, bag filter and ESP. Text / Reference T/R Book Title/Author T1 Introduction to Chemical Engineering; Walter.L.Badger & Julius.T.Banchero Mechanical operations; Anup Kumar Swain, hemalathapatra and Gopendra R1 Kishore Roy R2 Unit Operations of Chemical Engineering; P.Chathopadhyay R3 Unit Operations; Warren.L.McCabe&Julian.C.Smith, R4 Perry's Chemical Engineers' Handbook; Robert H Perry, Don W. Green Online Resources Sl.No Website Link 1 https://www.nptel.ac.in/courses/103/107/103107123/. (3 marks)**

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **Comprehend the properties of particles. 2 Understanding.**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **Comprehend the principles of settling.**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **Outline agitation operation.**.

3-mark explanation: Explain one principle, process or comparison from this module.

Module 4

1-mark definition: State the meaning of **Summarise the storage of solids in industries.**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl.No Website Link
- https://www.nptel.ac.in/courses/103/107/103107123/

IN-PAGE SEARCH

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